Supplementary: Key Results Summary (v8)

Authors: Smart, Morgan Elliott; Smart, Jasper Keiron; Ara (Grok Bot AI collaborator).
Supplement to SCG-HMH: A Closed Derivative Chain for a Regenerative Nitrogen Plasma-Ring Generator, version 8 (29 October 2026; an erratum of v7, Zenodo 10.5281/zenodo.23291052;
licence CC BY 4.0). Every number here is taken from the paper; nothing is new.

Framing. These are paper predictions for a machine at the edge of its operational envelope. They are derived, not measured; nothing here is offered as experimental proof.

Booked baseline (published envelope)

Quantity	Value
Field, rotor speed, safety factor	5 T, 150,938 rpm, SF 2
Output per module	99.55 MW (128.60 MW at SF 1.5)
COP_th = P / (waste heat + electricity in)	0.2175
COP_e = P / electricity in (0.8 kW)	124,437
Waste heat that closes the books	457.66 MW per module
100-module train	9.9549 GW

Envelope points (derived, not booked)

Point	Output per module
13 T stator field, SF 2 (1,300 N YBCO bearings)	1,155.24 MW at 197,761 rpm
Back-EMF feedback, $g_0 = 0.10$, SF 2	5.565 T, 126.96 MW
Feedback runaway to the 20 T REBCO ceiling ($g_0 \geq 0.4727$, SF 2)	4,565.46 MW at 255,541 rpm
v7 worked ring (SF 1.07, above the continuous envelope)	141.22 MW at 200,000 rpm

Loss of field at speed is a fault handled by protection, not a design limit.

Scaling laws

- Ring equation: $P = \sigma(\omega r B)^2 K(1 - K) \cdot 2\pi r h \delta$ (Eq. 4).
- Per module, ranked (Table 13y): r^3 (+33.10 % for +10 %); B^2 and ω^2 (+21.00 %); then σ, h and δ (+10.00 %). r and ω are limited by rim stress, B by the magnet.
- Per site, the number of modules is set by the free waste heat available.

Staged-validation path

1. Bench gate: stable gap conductivity above 5 S/m and sustained net output above 50 W (the chain predicts about 50.4 kW at the gate conductivity).
2. Step-ups in chamber pressure, rotor speed and field, each proven before the next.

The full envelope is unlocked by small, incremental steps, each a known-good stage.